

Hydrogen sulfide induces direct radical-associated DNA damage.

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Hydrogen sulfide (H₂S) is produced by indigenous sulfate-reducing bacteria in the large intestine and represents an environmental insult to the colonic epithelium. Clinical studies have linked the presence of either sulfate-reducing bacteria or H₂S in the colon with chronic disorders such as ulcerative colitis and colorectal cancer, although at this point, the evidence is circumstantial and underlying mechanisms remain undefined. We showed previously that sulfide at concentrations similar to those found in the human colon induced genomic DNA damage in mammalian cells. The present study addressed the nature of the DNA damage by determining if sulfide is directly genotoxic or if genotoxicity requires cellular metabolism. We also questioned if sulfide genotoxicity is mediated by free radicals and if DNA base oxidation is involved. Naked nuclei from untreated Chinese hamster ovary cells were treated with sulfide; DNA damage was induced by concentrations as low as 1 micromol/L. This damage was effectively quenched by cotreatment with butylhydroxyanisole. Furthermore, sulfide treatment increased the number of oxidized bases recognized by formamidopyrimidine [fapy]-DNA glycosylase. These results confirm the genotoxicity of sulfide and strongly implicate that this genotoxicity is mediated by free radicals. These observations highlight the possible role of sulfide as an environmental insult that, given a predisposing genetic background, may lead to genomic instability or the cumulative mutations characteristic of colorectal cancer.